

Introduction to Web Technologies — Assignment 1

Internet & WWW basics · Structure of a web page · Basic HTML document · Semantic tags, tables, forms, div & span

Step 0. Before you write any code

Your final exam is a project that you build across the whole semester, starting with this assignment. So first choose your theme, then your team. A team is a maximum of three students. Working individually is equally welcome and is graded on the same scale; only the number of pages changes. Team and theme are fixed for the semester: every week you extend the same repository, you never start over.

Theme

Choose a real, physically existing organization or place in your city that you can visit or contact in person—a café, gym, student club, barbershop, clinic, shop, animal shelter, driving school, museum, workshop. Fictional or generic companies are rejected. No two projects in the group may take the same organization; first come, first served.

What to build

index.html, colophon.html (how the site was made) and two more pages per student - so four pages if you work alone, six for two students, eight for three. Each page holds 150–250 words of your own text and the author's name in an HTML comment and in a meta author tag. **The site is a set of local files opened directly in your browser: you do not buy a domain and you do not host anything.** Hosting and deployment come later in the semester.

Required on every page

Doctype, lang, charset, viewport, a description and author meta tag, a unique title, exactly one h1 and correct heading order without skipped levels. Every page is built on a semantic skeleton: header with the site name, nav containing a list of relative links to every page of the site, main for the content, footer with contacts and a copyright entity. Add two comments that explain why you did something rather than what the tag is.

Required across your own two pages

Structure: section, article, aside, figure with figcaption. At least one table describing real data, with caption, thead, tbody and th with scope. Lists: a nested list, an ordered list with an attribute, a definition list. Links: an external link with target and rel, a mailto link, a tel link, two links to an id on the same page. Three images with meaningful alt. Text: strong, em, b, i, mark, small, sub or sup, two abbr with title, a blockquote holding a real quote you collected yourself, an inline q, cite, code, pre, kbd, samp, hr, br and four different HTML entities. The code, pre, kbd and samp tags belong naturally in colophon.html.

div and span

Use each of them at least once, but only where no semantic tag fits, and put a comment next to it explaining why nothing else worked. **A div used where header, nav, main, section, article or footer belongs costs 5 points each time.**

Forms

Each student writes at least one form on one of their own pages, relevant to the project - booking, feedback, registration, order. It must contain: method and action attributes, fieldset with legend, labels connected to their fields by id, and the input types text, email, tel, number and date, plus a radio group, a checkbox, a select with options, a textarea, the required and placeholder attributes, a submit button and a reset button. **The form does not send anything yet- there is no server behind it** - and you must say so in a comment inside the form. Server-side handling comes later in the course.

Forbidden

No CSS in any form, no JavaScript, no frameworks, templates or site builders. Your pages must be unstyled- styling starts next week and it is much easier on clean markup. Also no Lorem ipsum, no placeholder text, no stock or AI-generated images.

Tag checklist

Submit a checklist file naming every required tag together with the file and the line number where you used it, and which student wrote it. Missing or wrong line numbers mean the item is not counted.

Task A — Anatomy of a real page

Pick a real website similar to your project and study its source. Record its doctype, lang, charset, viewport, title and three meta tags. Note which semantic tags it uses, where it uses div instead, and how its forms and tables are built. Draw its page structure by hand on paper — signed, dated and photographed. Then name three structural mistakes you found in it and explain how you avoided each one in your own pages.

Task B — Written part

In 150–200 words of your own writing, explain what a web page is made of and how a browser turns your HTML file into what you see on screen. Then name three specific places in your own files where you chose a semantic tag over a div and say why, and explain what happens when a visitor presses your submit button today. A general textbook answer with no reference to your own files scores zero.

Submission

A Git repository containing your pages, an images folder, a PDF report of both tasks with all screenshots, the tag checklist, the hand-drawn sketch, an AI log and a README. Everyone commits from their own account: at least four commits across three different days, six if you work alone. Every page must pass the W3C validator with zero errors. All content must be real — real prices, opening hours, address and phone, at least one real quote with a name and date, and at least three photographs taken by you.

Defense in class

I will ask you to explain random lines of your code, to explain a page you did not write, to justify why you chose one tag instead of another, and to make a small live edit such as adding a row to your table or a field to your form. Your grade is the project score multiplied by how well you explain your own work; if you cannot explain it, the score is zero. Work that is submitted but not defended is not evaluated, even if it was uploaded before the deadline. The deadline gives you the right to be graded; the defense gives you the grade.

Grading (100 points)

Theme and evidence of contact 8 · Task A 15 · Task B 10 · semantic structure and tables 20 · forms 20 · tag checklist 15 · authenticity of content 7 · process and commit history 5. Penalties: 10 for any CSS or JavaScript, 5 per div used in place of a semantic tag, 5 per validator error, 5 per broken link, 15 for generic or placeholder content.

AI policy

You may ask AI to explain concepts. You may not use it to write your pages, your text, your report or your images, and every question you ask must be recorded in the AI log. The parts of this assignment that carry the most points - your own photographs, real data from a real place, your hand-drawn sketch, your commit history and the live defense cannot be produced by AI.